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THEORETICAL ARTICLE

Relation between digital innovation and industry 4.0: existing themes and emerging research trends

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Abstract

The primary objective of the research study is to investigate the influence of innovative digital tools and technologies in attaining the fundamental aspects of Industry 4.0, including automation, integration, traceability, flexibility, safety, and security. This study utilizes a systematic literature review, employing keyword search criteria to examine 187 relevant articles and evaluate the implementation and consequences of technologies, such as artificial intelligence, big data, cyber-physical systems, cloud computing, and digital twin, across a variety of Indian economic sectors, including retail, healthcare, manufacturing, agriculture, entertainment, and e-commerce. The study's results indicate that digital technologies can improve parameters such as automation, transparency, integration, traceability, flexibility, safety, and security, while promoting sustainable development through reduced waste, operational costs, and time, and optimum resource utilization. The insights from this study will be helpful for future inquiries and economic initiatives, encouraging the adoption of digital innovations in fields such as women's empowerment, public administration, and skill development.

Keywords Digital innovation · Digital technologies · Industry 4.0 · Sustainable development

1 Introduction

To manage complexity, new technologies are essential in our machine and technology-driven era, with the Fourth Industrial Revolution (Industry 4.0) transforming human life [124, 87]. It also changes how innovation is currently managed [70]. The first industrial revolution, starting in the mid-eighteenth century, mechanised industry and transformed global economies. The second industrial revolution significantly advanced mass manufacturing. The third revolution saw widespread adoption of information technology and computers. In contrast, Industry 4.0 trends towards automation, machine-to-machine communication, and cyber-physical systems to

Extended author information available on the last page of the article

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reduce human involvement and increase productivity [80]. These technologies are crucial for digital innovation today [70].

Klaus Schwab, executive chairman of the World Economic Forum, first coined "Industry 4.0" [119], describing it as an advanced version of the third industrial revolution, where machines self-analyse [63]. Industry 4.0 encompasses artificial intelligence, blockchain, cloud computing, data analytics, information and communication technology (ICT), enterprise architecture, the Internet of Things (IoT), cyber-physical systems, and enterprise integration [15]. These technologies are essential for maintaining a stable industrial ecosystem in this century. Due to its rapid growth, researchers have shown significant interest in the field [97].

Digitally innovated technologies enhance organizational functions by achieving speed, accuracy, integrity, transparency, accessibility, flexibility, and security. AI and ML, for example, facilitate predictive maintenance, process optimization, and automated decision-making, reducing human intervention and allowing human resources to focus on strategic planning and business expansion [69]. These technologies integrate mechanical processes with digital control systems, enabling real-time monitoring and autonomous control of manufacturing processes. IoT simplifies data collection from various sources and enables automated real-time responses, ensuring transparency and accuracy for stakeholders [86]. Organizations can analyze vast amounts of data through big data analytics, providing insights into operations and enhancing understanding via performance metrics [35]. Predictive analytics foresee and mitigate risks, improving workplace safety. Cloud computing offers a unified platform where different systems and applications can communicate and share data, adaptable to the organization's needs, addressing the increasing risk of cyber theft and technological change [154].

These digital technologies equip organizations to progress toward Industry 4.0 by enhancing automation, transparency, integration, traceability, flexibility, safety, and security, while also fostering sustainable development through greater efficiency, resource management, and environmental awareness. Industry 4.0, no longer novel to corporations, originated in Europe, particularly Germany, and has rapidly globalized [94, 104]

. In the digital and globalized age, little remains unknown. Consequently, Industry 4.0 has become a buzzword among corporations and academics. Numerous studies now focus on various economic sectors, especially manufacturing, supply chain management, and service industries. Industry 4.0 influences both economic sectors and market demands. Thus, companies must prioritize their research and development departments to lead in digitally innovative business strategies [50, 54].

Business organizations recognize that digital innovation is essential for sustaining Industry 4.0's vitality, as it allows businesses to achieve Industry 4.0 objectives and facilitates societal transformation through digitally innovated products, processes, and services [24]. Consequently, this research study systematically reviews previous literature to identify current trends in Industry 4.0 and the role of digital innovation. It aims to elucidate the relationship between digital innovation and Industry 4.0 by evaluating relevant literature and examining the impact of digital innovation across various economic sectors. The study's findings help identify challenges organizations face in implementing this technology, providing avenues for future research. A combination of research techniques, including systematic literature review (SLR) and document analysis, is employed to assess digital innovation's impact on the sustainable development of the Indian economy while achieving Industry 4.0 dimensions. This research study addresses the following questions:

RQ1: How "Industry 4.0" and "Digital Innovation" are interrelated with each other?

RQ2: What has been investigated by previous research studies on "Industry 4.0" and "Digital Innovation"?

RQ3: What are all the research gaps, limitations, and recommendations for future research scholars and practitioners in the fields of "Industry 4.0" and "Digital Innovation"?

To address the study concerns, we systematically evaluated over 187 publications from 2011 to 2021 within the Web of Science database, serving as a repository of prior knowledge [8]. For RQ1, we conducted a descriptive analysis to identify common keywords, industries applying this subject, countries publishing these studies, and the technologies sustaining Industry 4.0. Addressing RQ2 involved reviewing prior literature based on the number of related journals, authors publishing on the theme, industries where related studies or digital innovation are practiced, and the adoption of Industry 4.0 and Digital Innovation in developing and developed economies. RQ3 was addressed by analyzing the literature to determine the impact of digital innovations on sustainable economic development and identifying future research opportunities for scientists, researchers, economists, entrepreneurs, and academicians.

This research study is organized into six sections. Section 1 is the introduction. Section 2 addresses RQ1 and discusses the importance of digital innovations for achieving Industry 4.0 dimensions for a sustainable economy. Section 3 covers RQ2, detailing the review process and previous research on Industry 4.0 and digital innovation. Section 5 addresses RQ3 by summarizing the previous section and defining the study's various themes in the literature. Section 6 considers the managerial implications of the study. Finally, Sect. 7 concludes by discussing the research limitations.

2 Theoretical contribution

This section, divided into three subsections, elaborates on digital innovation, Industry 4.0, and their interrelationship.

2.1 Digital innovation (DI)

Digital innovation (DI) offers numerous benefits to individuals and organisations, including aiding routine tasks, optimizing administrative and corporate processes, and transforming complex decisions into strategic ones [12, 47]. DI facilitates the modernization of organisational activities through various products and processes

[51, 167]. By integrating technologies such as IoT, cybersecurity, cloud computing, blockchain, and machine learning, organisations can automate, integrate, and secure their operations for enhanced efficiency [39].

DI addresses local demands and removes institutional barriers to achieve Industry 4.0 dimensions, fostering awareness and technical support for new transition strategies. It promotes synergies among processes and functions, encouraging collaboration and further innovation. DI involves all stakeholders in the transition to sustainability through Industry 4.0 implementation.

Using advancements like AI, ML, and big data analytics, businesses can enhance consumer experiences by creating tailored products and services [1]. Companies like Netflix, Google Pay, Uber, and Amazon have improved consumer experience through home entertainment, cashless transactions, seamless trips, and doorstep delivery of essentials. To thrive in this competitive era, enterprises leverage digital innovation to manage stakeholders and achieve Industry 4.0 goals [46]. Success in Industry 4.0 requires transforming factories into "smart factories," upgrading supply chains, enhancing customer service, and continuously innovating products with digital innovation [101].

2.2 Industry 4.0

Industry 4.0 marks the fourth industrial revolution, distinguished by sophisticated management and control of the product life cycle to accommodate increasingly personalized customer demands [117], Thoben, Wiesner, and Wuest, [25]. Its primary goal is to satisfy individual customer needs, impacting areas such as order management, R&D, manufacturing, recycling, and delivery [99]. This revolution transforms manual efforts into digitally mechanized systems, creating smart factories. It encompasses nine key digital technologies: simulation, big data analytics, autonomous robots, horizontal and vertical integration, augmented reality, additive manufacturing, industrial IoT, cloud computing, and cyber security. Industry 4.0 envisions an integrated, automated industrial production system to optimize processes.

Industry 4.0, as termed by the German government, aims to transform manufacturing into a smart industry [135]. It involves using ICT and innovative technologies to enhance various economic sectors, including manufacturing, services, banking, healthcare, and management [137]. This initiative encourages the adoption of advanced technologies like IoT, Cloud computing, Big Data, Cyber Security, and Blockchain to achieve standards such as simulation, flexibility, security, autonomy, integration, and improved decision-making [9]. Industry 4.0 focuses on technological advancements, agile processes, and essential business operations within a complex environment, integrating machines and humans in a communication network that ensures mobility, flexibility, security, accessibility, and vertical and horizontal integration [7].

With technological advancements, Industry 4.0 has boosted productivity, revenue, job creation, and investments. Despite emphasizing automation, integration, virtualization, and digitalization, it also significantly impacts the human-worker interface in smart factories. It promotes connecting physical items like sensors, devices, and enterprise assets to each other and the Internet [125]. Industry 4.0 transforms regular machines into smart, self-aware, self-reliant, and self-learning entities, enhancing overall efficiency, performance, and productivity.

2.3 Relation between digital innovation and industry 4.0

Industry 4.0 has transformed business trends, compelling firms to modernize their process models and enhance operations with digital innovation tools. Research highlights the need for companies to prioritize their supply chain, production processes, and customer service to stay aligned with Industry 4.0 [10, 16]. Achieving Industry 4.0 goals without integrating digital innovation into organizational procedures is difficult. Digital innovation presents numerous opportunities to optimize practices or processes [30, 51], such as automating manufacturing to save time and reduce costs [39]. It also improves customer service through the use of customer databases and knowledge management systems [1]. To tackle Industry 4.0 challenges, companies must adopt smart manufacturing systems, streamline supply chains, enhance customer service, and develop innovative products, all of which can be achieved through digital innovation [101].

Digital innovation and Industry 4.0 together improve business processes by introducing digital technologies and dimensions like online learning platforms, app-based medicine delivery, health consultations, and home gym facilities, transforming business models for sustainability [47]. Industry 4.0 has transformed business models, digital technology strategies, production and operations, logistics, SCM systems, and human resource skills. Studies indicate that digital innovation helps companies gain a competitive edge by creating value, ensuring strategic fit, and enhancing agility [34, 43, 84, 90, 108, 120, 138].

Adopting digital technology in an organization requires first defining its purpose or identifying the organizational culture [116]. Before implementing digital innovations (DI) in business processes, organizations must adjust existing employees, technology, and systems [66]. The introduction of any digital technology should not disrupt organizational principles, practices, or procedures. Examining organizational and employee behavior is essential to prevent resistance [161]. Research indicates that DI influences innovation performance based on available information and data [138]. Additionally, DI is effective only if the organizational culture aligns with the employees' goals and vision [126, 130].

3 Research methodology

This process involves several research questions and their resolutions, such as selecting the research field, choosing the data source, determining the research method, and justifying the research's necessity. Our focus will be on Industry 4.0, a vast and complex subject, necessitating a specific focus. Thus, we chose Industry 4.0 in relation to Digital Innovation (DI), an emerging and fast-evolving research area.

To address the research questions, we conducted a thorough literature review on Industry 4.0 and DI. This review helped assess the credibility and validity of existing research [136]. A thorough review and analysis of the literature add to the existing body of knowledge [31, 136]. The phases of this literature review are outlined in Fig. 1 below:

3.1 Material collection

A structured keyword search of the ISI Web of Science (WoS) database was conducted to encompass all aspects of digital innovation and Industry 4.0. Compared to Emerald, Springer, IEEE, Elsevier, Taylor & Francis, and others, WoS is a leader in authentic, ethical publications and is renowned for its extensive coverage of high-impact journals in English [3]. Industry 4.0 emerged post-2010, later extended by Industry 5.0, an advanced version of Industry 4.0 [156]. Therefore, the search included articles published between 2011 and 2021. Search terms included "digital innovation," "digital technology," "digital transformation," "digitalization," combined with "Industry 4.0," "smart factory," "smart production," "smart manufacturing," and "fourth industrial revolution" using Boolean AND. This ensured that all selected articles (n = 490) had an established connection between DI and I4.0. After reviewing the abstracts, articles were narrowed to 251 depending upon their relevance. Further refinement excluded those addressing non-manufacturing aspects of Industry 4.0 and digital innovation (Energy Industry, Marketing, economy-based studies, actor-network theory, etc.), resulting in 187 articles. The final evaluation involved a thorough review of these 187 articles.

Additionally, research articles on Industry 4.0 and DI cited within these 187 publications were reviewed to enhance subject comprehension. This snowball method ensured that essential knowledge from articles missed in the initial search was captured. The following sections present the distribution of selected articles by journal, with the selection approach illustrated in Fig. 2.

3.1.1 Research identification

The initial step involves defining the research material for inclusion in the manuscript. After thorough review, only relevant and English-language research articles were selected. The stages of the SLR are illustrated in Fig. 2.

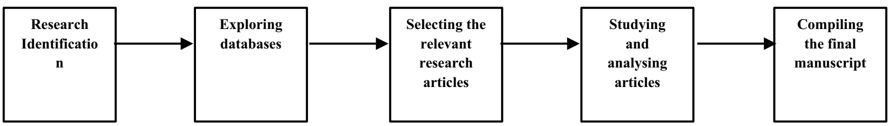

Fig. 1 A process model of a systematic review of the literature (Source: [17]

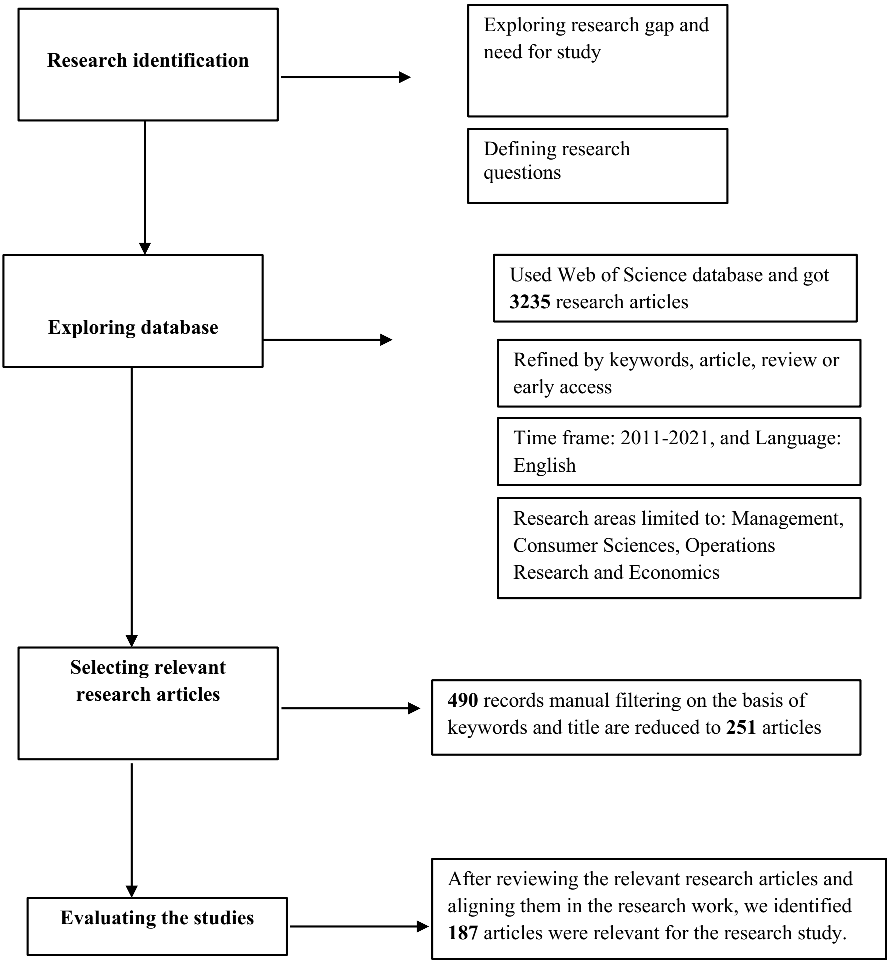

Fig. 2 Expanded flow chart of systematic literature review

3.1.2 Exploring databases

The study utilized the Web of Science database, known for its extensive collection of high-quality articles [145]. The literature review followed the flow chart in Fig. 1. The search commenced in January 2021, initially identifying 3235 publications.

3.1.3 Selecting the relevant research articles

The research articles were sourced exclusively from the Web of Science database using search terms like 'Digital Innovation,' 'Digital Technology,' 'Digital transformation,' 'Digitalization,' 'Industry 4.0,' 'Smart Factory,' 'Smart production,' 'Smart Manufacturing,' and 'Fourth Industrial Revolution.' The search covered publications from 2011 to 2021, yielding 490 articles. Filters were organized based on the researchers' fields of interest, including management, consumer sciences, operations research, and economics.

3.1.4 Evaluating and analysing the studies

For this research, only journals and review articles with a major impact on those fields were used as sources. After 3235 research publications at the first level, the number of articles was reduced to 490.

3.1.5 Combining the research work

In 2011, DI and Industry 4.0 emphasized research [53, 94, 104]. The primary indexes for this study were SCI-Expanded, SSCI, A & HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, and ESCI. Publications were selected based on keywords, abstract, and conclusion. After filtering, 251 research publications were identified for a systematic literature review (SLR). From these, 187 studies were deemed relevant and used for this research after abstract analysis.

3.2 Descriptive analysis

This section analyzes the articles descriptively. There has been a steady rise in publications on Industry 4.0 and DI. The implementation of DI across various sectors, such as manufacturing, medical, agriculture, hospitality, tourism, banking, finance, and education, has led to a significant increase in research studies due to heightened academic interest. Recent literature reviews confirm this upward trend [5, 14, 96], indicating growing interest among researchers and scientists in Industry 4.0 and DI.

Table 1 Annual Scientific Production

Year	Articles
2011	1
2014	2
2016	6
2017	4
2018	38
2019	47
2020	85
2021	4

3.2.1 Year-wise publication details

Table 1 displays the number of articles published on I4.0 and DI, indicating an escalation of scholarly attention since 2011 when nations like the United States, Germany, and China announced the fourth industrial revolution [53, 94, 104, 157]. The 96 selected articles, published across 20 journals, highlight the topic's extensive coverage and interest from high-impact journals. The geographic distribution of first authors reveals that most publications originate from China, the United States, and Italy, followed by the United Kingdom, Germany, and Brazil.

3.2.2 Number of articles based on the type of research methods

The authors divided the research articles into two main categories: reviews and empirical or case-based studies. The empirical and case-based articles focus on integrating digital innovations like blockchain, big data analytics, cloud computing, cyber-physical systems, and IoT, technologies into manufacturing to accomplish I4.0 goals and objectives. The review articles discuss the rise of Industry 4.0, its parameters, implementation challenges, potential theoretical models, and a conceptual framework for Industry 4.0 implementation. Among the articles, 96 (51.33%) employed an empirical or case study method, while 91 (48.67%) used a review strategy.

3.3 Review classification framework

The category selection aims to create a conceptual framework for the research. Four key dimensions addressed the study's research issues: keyword analysis, country performance, relevant research sources, and author performance. The study examined emerging digital technologies transforming manufacturing firms into smart factories under Industry 4.0 to achieve sustainable competitive advantage.

4 Review findings on digital innovation and industry 4.0

This section divides the findings and analyses into two subsections, each addressing one of the research questions introduced earlier: (1) RQ1: How are "Industry 4.0" and "Digital Innovation" related? and (2) RQ2: What have past research studies on "Industry 4.0" and "Digital Innovation" examined? The third research question (RQ3) is discussed in the paper's discussion section. Each part reports the analysis outcomes. Initially, descriptive information about the sample is provided based on relevant research sources, followed by tables and a detailed analysis.

4.1 Data based interrelationship between DI and I4.0

In this section, authors used keyword analysis, cluster analysis, and inter-country research collaboration to explain the interrelationship between digital innovation and industry 4.0.

4.1.1 Keyword analysis

A keyword analysis identified the link between Industry 4.0 and Digital Innovation (DI). Table 2 lists the most frequently used author keywords. "Smart manufacturing" was the primary keyword with 92 occurrences, integrating Industry 4.0 and Digital Innovation. “Industry 4.0” appeared 89 times, making it the second most common term. Other frequently used keywords include Industry 4 (54), Smart Factory (47), Digital Transformation (28), Manufacturing (28), Internet of Things (25), Big Data (16), Digital Twin (15), Digitalization (13), Fourth Industrial Revolution (13), and Cyber-Physical Systems (12). The “smart factory” term, used to describe the link between “Industry 4.0” and “Digital Innovation,” refers to a fully automated production system with minimal human intervention, fewer errors, quick response, and transparency. Figure 3 and Table 2 illustrate this with a word cloud and word frequency count.

The most frequent word in Table 2 is "example." Table 2, covering various areas, shows strong connections to Industry 4.0 and digital innovation. Notable terms include "industry 4.0," "smart factory," "manufacturing," and "digital transformation." Additionally, terms like "internet of things (IoT)," "digital twin," "cyber-physical systems," and "big data" are commonly used by authors. As shown in Table 2, Industry 4.0 and digital innovation are primarily applied to industrial sectors.

Most research articles focused on Industry 4.0’s impact on manufacturing operations. Studies explored opportunities and challenges in production, supply chain, and business functions. However, only a few papers discussed effects on healthcare, education, or other sectors. More research is needed to assess Industry 4.0’s influence in diverse domains.

4.1.2 Cluster analysis

The data analysis revealed that each digital technology served a distinct purpose: cyber-physical systems for virtualization, IIoT for enterprise resource planning, smart manufacturing, technological innovation in production and operations

Table 2 Most Frequent Word

Words	Occurrences	Words	Occurrences
Smart manufacturing	92	Digitalization	13
Industry 4 0	89	Fourth industrial revolution	13
Industry 4	54	Cyber-physical systems	12
Smart factory	47	Blockchain	11
Digital transformation	28	Machine learning	11
Manufacturing	28	Systems	9
Internet of things	25	Cloud manufacturing	8
Big data	16	Cyber-physical system	8
Smart	16	Digital technology	8
Digital twin	15		

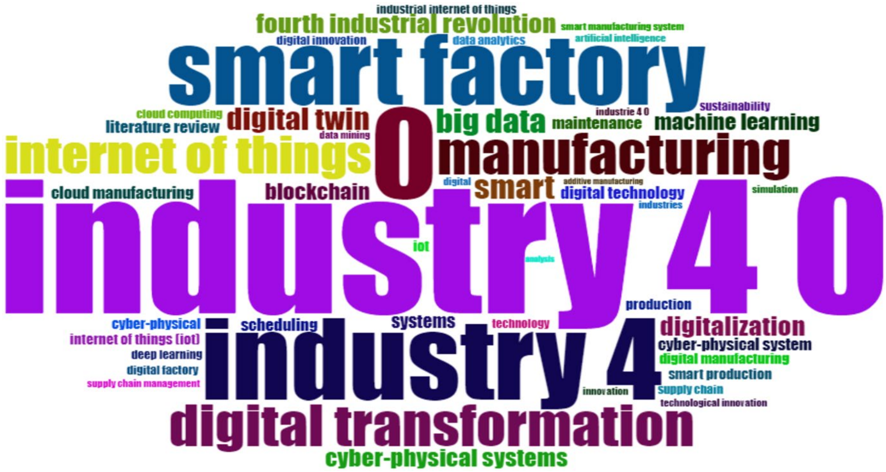

Fig. 3 Word cloud based on the authors keywords

management, and digital technologies for smart factories. Some technologies are integrated to boost performance, such as combining cyber-physical systems with the Internet of Things or blockchain technology to achieve smart factories, smart manufacturing, and smart industries. Researchers identified these keywords as specific clusters within Industry 4.0. Figure 4 illustrates the cluster analysis of Digital Innovation and Industry 4.0 research.

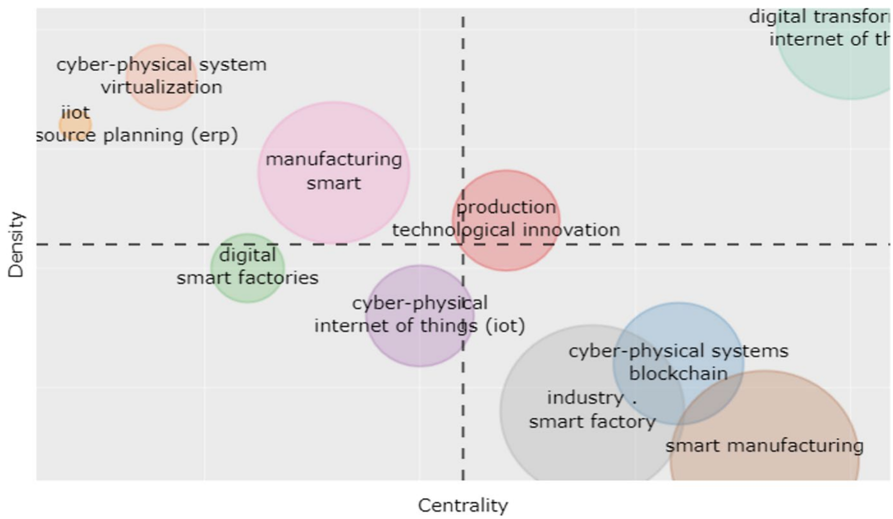

Fig. 4 Cluster analysis of the Digital Innovation and Industry 4.0 research

4.1.3 Relevant sources for research

Publishing trends highlight the growing interest of scholars, scientists, and academicians in I4.0 and DI research. Major journals in this field include Technological Forecasting and Social Change, International Journal of Production Research, Journal of Manufacturing Systems, International Journal of Advanced Manufacturing Technology, and Computers & Industrial Engineering. Table 3 summarizes the top twenty literature sources utilized in this research.

Technological Forecasting and Social Change, with 49 articles, is the leading journal. Research demonstrated the impact of Industry 4.0-based digital technologies on controlling air pollution in the U.S., exacerbated by tourism [2]. Another study examined the unplanned expansion of private universities in residential areas, assessing their role in the era of Industry 4.0 [4]. Cuomo et al. [26] evaluated the impact of big data analytics on enhancing customer experience by extracting insights from customer feedback.

The International Journal of Production Research, with 40 articles, is the second-most authoritative source. Ding et al. [32] reviewed how smart manufacturing is central to Industry 4.0, detailing the implementation of cyber-physical systems and twin technologies for real-time monitoring, communication, and operations control. Gunasekaran et al. [45] reviewed the literature on the evolution, attributes, and drivers of agile manufacturing for future researchers.

Table 3 Journals publishing the highest number of articles

Journals	Rank	Freq
Technological Forecasting and Social Change	1	49
International Journal of Production Research	2	40
Journal of Manufacturing Systems	3	37
International Journal of Advanced Manufacturing Technology	4	35
Computers & Industrial Engineering	5	25
IEEE Transactions on Industrial Informatics	6	25
International Journal of Computer Integrated Manufacturing	7	21
International Journal of Production Economics	8	19
Production Planning & Control	9	18
Computers in Industry	10	16
Journal of Intelligent Manufacturing	11	12
Enterprise Information Systems	12	10
Industrial Marketing Management	13	10
Industrial Management & Data Systems	14	7
Applied Soft Computing	15	4
IEEE Transactions on Engineering Management	16	4
Annals of Operations Research	17	3
Benchmarking-An International Journal	18	3
Expert Systems with Applications	19	2
Journal of Retailing and Consumer Services	20	1

Lu and Asghar [83] introduced semantic-aware cyber-physical systems (CPS) in the Journal of Manufacturing Systems, facilitating machine-to-machine communication using IIoT and CPS to establish a modern architecture.

Seven journals were identified as having relevant publications:

- **Technological Forecasting and Social Change**: Focuses on technological forecasting and future studies as planning tools considering social, environmental, and technological factors, with 49 articles.
- **International Journal of Production Research**: Emphasizes industrial engineering, management science, and operations research to advance operations management and logistics education and research, with 40 articles.
- **Journal of Manufacturing Systems**: Explores theories, techniques, and transdisciplinary research to innovate and improve manufacturing systems, with 37 articles.
- **International Journal of Advanced Manufacturing Technology**: Covers issues and research on manufacturing technology across industries, focusing on manufacturing processes, machines, and process integration, with 35 articles.
- **Computers and Industrial Engineering**: Discusses new computerised methodologies for solving industrial engineering problems, with 25 articles.

4.1.4 Inter-country research

Following the announcement of Industry 4.0, economies must adopt digitally innovative technologies to transform traditional manufacturing into smart factories and maintain global competitiveness. Scientists and researchers conducted inter-country studies to explain the adoption and implementation of digital technologies in factories. Table 4 lists the top twenty associated author countries, showing the country of origin based on the number of collaboratively published documents. China, the United States, and Italy are the top three countries in article production according to Table 4. Publication frequency was identified based on the performance of countries with SCP (single country publishing), MCP (multiple country publication), or no articles. China (82 articles) had 46 SCP and 36 MCP, the USA (35 articles) had 27 SCP and 8 MCP, and Italy (32 articles) had 26 SCP and 6 MCP.

Table 4 highlights the interest and ambition of author countries during the fourth industrial revolution, an extension of the third industrial revolution aimed at automating manufacturing to optimize processes and achieve sustainability. This advancement, termed "Reindustrialization and Industrial Internet" by the US government, "Industry 4.0" by the German government, and part of the national policy "Made in China 2025" by China, reflects global efforts to lead this technological shift [76].

This transformation will automate manufacturing processes like production planning, job scheduling, monitoring and control, logistics, and supply chain management using innovative digital technologies such as the "Internet of Things," "Big Data Analytics," "Digital Twin," "Machine Learning," and "Cyber-Physical Systems" [40, 60, 82, 102, 131]. Table 5 details the application of these technologies to various production processes.

Table 4 Corresponding Author's Country

Country	Articles	Freq	SCP	MCP	MCP_Ratio
China	82	0.23977	46	36	0.439
USA	35	0.10234	27	8	0.229
Italy	32	0.09357	26	6	0.188
United Kingdom	26	0.07602	16	10	0.385
Germany	20	0.05848	14	6	0.3
Brazil	16	0.04678	9	7	0.438
Korea	14	0.04094	10	4	0.286
Spain	14	0.04094	12	2	0.143
France	13	0.03801	4	9	0.692
Sweden	9	0.02632	4	5	0.556
Australia	7	0.02047	4	3	0.429
Netherlands	6	0.01754	4	2	0.333
India	5	0.01462	5	0	0
Turkey	5	0.01462	5	0	0
Finland	4	0.0117	2	2	0.5
Norway	4	0.0117	3	1	0.25
Denmark	3	0.00877	2	1	0.333
Greece	3	0.00877	2	1	0.333
Hungary	3	0.00877	3	0	0
New Zealand	3	0.00877	2	1	0.333

4.1.5 Authors' contribution

The contributions of authors in research publications are essential. Tao F is the most referenced, utilizing cloud computing to enhance big data analytics, cyber-physical systems, and digital twins [20, 22, 131, 132, 133]. Other frequently cited authors include Dmitry Ivanov, Henning Kagermann, Ray Zhong, and Jay Lee. Author productivity is summarized by the total publications of the top 20 cited authors. Table 6 classifies the highly cited authors for DI and I4.0.

Most research targets the manufacturing sector. Other industries, such as agriculture, healthcare, education, cryptocurrency, services, energy, transportation, FDI, sales and management, and hospitality, are still navigating the DI and I4.0 landscape [118, 123, 164, 166], Nojd et al., 2020; [4].

4.2 Previous research studies on "Industry 4.0" and "Digital Innovation"

Industry 4.0 represents a smart, efficient, safer, and ecologically sustainable scenario where items are personalized and production is automated through digital technologies, such as the Internet of Things (IoT), Big Data Analytics, Cyber-Physical Systems, Machine Learning, and Cloud Computing [127]. It combines ICT (Information and Communication Technology), industrial technology, and management

Table 5 Digital technologies used in manufacturing processes for sustainable competitive advantage

S. no	Themes	Procedures of business organizations	Author/year
1	IoT	Efficiency framework for manufacturing industries, instantaneous coordination, networked machine integration	[59], Corradi et al., [63, 64, 117, 118, 128, 64, 67, 103]
2	Big data analytics	Enhancing system efficiency; ideal route optimization; customer relationship management	[81, 167, 75, 27, 123, 163]
3	Digital twin	Managing manufacturing location; enhancing and aligning both the physical and virtual layers within the manufacturing facility	[21], Damjanovic & Behrendt, [134, 148, 169, 139]
4	Machine learning (AI)	Production processes, Just-in-time scheduling, factory floor management; Resource allocation planning; radio wave frequency (RF)/visible light communication (VLC) industrial network framework; Enhancing system performance;	[144, 78, 151, 40, 142, 74, 73]
5	Cyber-physical systems	Supply Chain Planning, addressing various scheduling issues; Managing reusable transit items, Machine-to-machine (M2M) coordination; Resource tracking and allocation on the shop floor; Dynamic closed-loop scheduling system	[60], Jung et at. [142], Rossit etal., [143, 83, 56]
6	Cloud computing	Implementation of a hierarchical fog server within the firm's network; Task delegation, route-map for factory outlet; Developing an ERP (Enterprise resource planning) system; Cloud-based Kanban decision support system	[18, 113, 85, 154, 121]
7	Blockchain technology	Blockchain architecture to reshape traditional IIoT and for other digital devices; To assess the influence of Bitcoin price (BP) on oil price (OP) and vice versa;	[58, 57, 143, 128, 171]

Table 6 Most cited authors for DI and I4.0

Authors	Citations	Authors	Citations
Tao F	131	Wang Sy	60
Ivanov D	97	Monostori L	59
Kagermann H	95	Liao Yx	56
Zhong Ry	78	Mourtzis D	52
Lee J	77	ChienCf	51
Lee Jay	77	Wan Jf	48
Xu Ld	72	Gunasekaran A	42
Zhang Yf	72	Tortorella GL	42
Kusiak A [71]	71	Hermann M	41
Wang Lh	62		

technology to achieve advanced production operations [76]. Consequently, this fourth industrial revolution has been termed "Reindustrialization" and "Industrial Internet," adopted by national strategies in China and Germany as "Made in China 2025" and "Industry 4.0," respectively.

Both developed (the United States, the United Kingdom, Japan, and Germany) and developing economies (China and India) have recognized and promoted Industry 4.0 [6]. A poll indicates that China leads in integrating digital technology into manufacturing, transforming them into "smart factories," with 2500 patents, surpassing the United States' 1065 and Germany's 441 [112]. This underscores the pivotal role of DI in enhancing industrial processes to achieve I4.0 dimensions.

Industry 4.0 is essential for manufacturers to address growing customer expectations, improve competitiveness, drive innovation, and enhance productivity. Efforts include conserving resources and energy, achieving financial and performance sustainability, supporting management decisions, utilizing advanced information-sharing technologies, innovating business models, digitalizing and integrating networks, and digitalizing products and services [38, 55]. To fully realize Industry 4.0, understanding its core features is essential, such as digitization, optimization, customization of manufacturing processes, automation, adaptation, human–machine interaction, value-added services, automatic data sharing, and communication [110, 79].

These components define I4.0 as the implementation of digital technology in manufacturing. Characteristics include automation, virtualization, integration, collaboration, interoperability, flexibility, safety, security, real-time availability, service orientation, and energy efficiency [55, 106]. The accompanying diagram illustrates how digital technologies transform traditional factories into smart factories by integrating these elements.

Digital innovation is essential for implementing and sustaining Industry 4.0 features. By enhancing supply chains, automating work environments, personalizing products, and developing smart products, digital technologies convert traditional manufacturing into smart factories [92]. A survey of 92 manufacturing enterprises

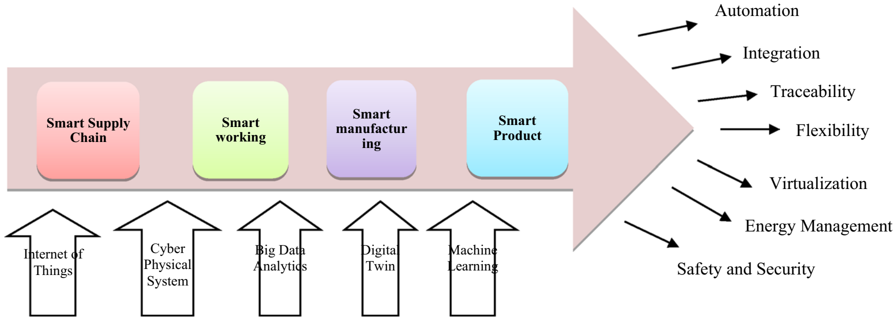

Fig. 5 Use of digital technologies (Solution components) for sustaining Industry 4.0

in Brazil's Machinery and Equipment sector examined the role of digital technologies in implementing Industry 4.0 patterns [41] (Fig. 5).

Review data indicates digital technologies are utilized in MNEs (Multinational Enterprises), but many SMEs (Small and Medium-Sized Enterprises) struggle with Industry 4.0 implementation [91]. SMEs need financial and technical resources, specialized products, ISO compliance, supportive organizational culture, strong leadership, and employee participation to adopt Industry 4.0. Barriers include:

Capturing and interpreting machine-generated data for effective decision-making.
Requiring new structures, processes, and technologies for data analysis.
Ensuring data safety and security due to digital technologies, IoT, CPS, and Big Data Analytics.
Overcoming technical incompatibilities with standards and interfaces of digital technologies (IoT, CPS, Big Data Analytics, AI, AR).
Workforce resistance due to job security concerns and redundancy fears.

4.2.1 Role of digital technologies in industry 4.0

Digital transformation, as depicted in Fig. 6, combines advanced technologies to revolutionize manufacturing. By integrating IoT, big data, cloud computing, and

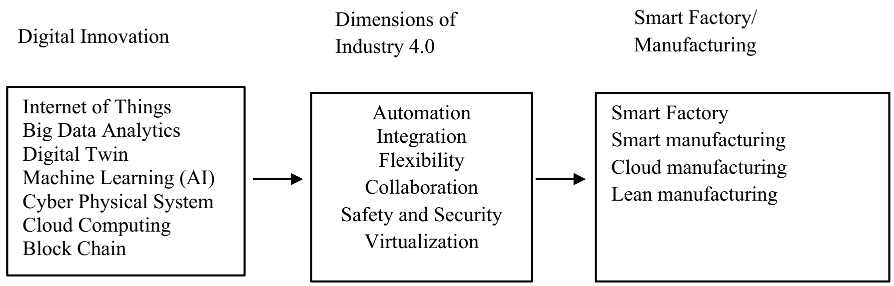

Fig. 6 Proposed conceptual framework

machine learning, industries can optimize their processes and adapt to an evolving market.

Research into emerging technologies is transforming the industry, promoting long-term growth. Clusters have emerged around digital innovation in Industry 4.0. We identified new clusters in key academic fields based on literature trends and gaps.

4.2.1.1 Internet of things (IoT) The Internet of Things (IoT) is a digital innovation extensively adopted by enterprises to realize Industry 4.0's goals. Customer-based IoT enhances customer experiences via smart products and services (e.g., smartphones, smartwatches, smart TVs, automatic washing machines, smart refrigerators). Manufacturing IoT improves factory flexibility, efficiency, productivity, and performance, turning it into a smart factory through smart objects, sensors, actuators, instruments, and software systems. Table 7 summarizes studies and emerging clusters related to the IoT.

The Internet of Things (IoT) is defined as 'a global framework for the information society that enhances services through the interconnection of physical and virtual entities via both established and emerging interoperable information and communication technologies' [95]. It enables connectivity by incorporating sensors and devices for the collection, transmission, and sharing of data [100], [102].

4.2.1.2 Big data analytics Big data technology manages large data volumes, characterized by volume, variety (structured, semi-structured, unstructured), and velocity (data processing ability) [19, 165]. Techniques like rule learning, cluster analysis, machine learning, natural language processing, pattern recognition, and spatial analysis are used in Industry 4.0 [49]. Table 8 summarizes studies and emerging clusters in Big Data Analytics.

Big data analytics manages data through various methods, including data modeling (unifying semantics), data integration (across product or process life cycles, both horizontally and vertically), data content, decision processing (monitoring conditions and optimizing networks), knowledge processing (generating models, adapting knowledge, transforming know-how, and real-time data access), and information security (network and data safety). To achieve Industry 4.0 objectives, information systems like Enterprise Resource Planning (ERP) and Customer Relationship Management (CRM) collect real-time data. Also, business firms collect outside information using OTT platforms, market analysis, industrial regulations, and competitors' financial reports to formulate strategies and take major decisions using on big data analytics. This enables firms to automate production control, forecast machine maintenance, detect and prevent issues, support operations, and allocate resources to other departments like R&D, selling and distribution, logistics, purchasing, and customer service. Twelve SAP consultants were interviewed to identify challenges in instigating big data solutions for smart manufacturing [75]. These challenges are categorized as organizational, personal, and technical.

Table 7 Internet of Things (IoT) based studies and emerging clusters

Industry 4.0 dimensions	Type of action taken	Digital technology	Industry	Country	Type of study	Author/year
	Performance model	IoT	Manufacturing	Korea	Empirical	[59]
Real-time scheduling and decision making	Production planning and scheduling	IoT	Automobile manufacturing	China	Empirical	[79]
Sensing	Manufacturing shop floor	IoT and Coloured Petri Net (CPN)	Engine manufacturing	China	Empirical	[164, 166]
Collaboration	Assembly Production	IoT, CPS, Virtual Reality (VR)	Electronics manufacturing	USA	Empirical	[68]
Virtualization	Smart interconnection to implement smart manufacturing	Industrial Internet of Things (IIoT) hubs	Manufacturing	China	Empirical	[71]
	Architecture for information service	IoT	Manufacturing	Korea,	Case-based study	[162]
	Design of ICT Platform	IoT, RAMI 4.0	Ice cream factory machines	Italy	Empirical	Corradi et al., [128]
	Bottleneck task dispatching method	IoT	Manufacturing	China	Experimental and descriptive	[58, 57]
Monitoring, tracking, and decision making	Connected micro smart factory	Digital twin, IIoT	Manufacturing	Korea	Case-based study	[102]
Sensor	Real-time tracking and monitoring system	IoT	Manufacturing	China	Explanatory and descriptive	[81]
	Interaction of digitalization and servitization	IoT	Manufacturing	Sweden	Empirical study	[64]
Security & privacy	Data security	IoT	Automobile Manufacturing	China	Exploratory and descriptive	[67]
	Monitoring and analyzing industrial systems	IIoT	Computer Science	Spain	Exploratory and descriptive	[141]

Table 7 (continued)

Industry 4.0 dimensions	Type of action taken	Digital technology	Industry	Country	Type of study	Author/year
Integration, automation	Creating cyber-physical production systems (CPPS) and linking functions across the entire product lifecycle	IoT	Manufacturing	Korea	Exploratory and case-based study	[103]

Table 8 Big Data Analytics based studies and emerging clusters

Industry 4.0 dimensions	Type of action taken	Digital technology	Industry	Country	Type of study	Author/year
Smart manufacturing Service	Reinforced learning, optimal routes planning	Big data analytics	Hotel industry	China	Exploratory and descriptive	[81]
Interoperability and traceability	Smart manufacturing decision making	Big data analytics	Manufacturing	USA	Case study	[167]
	Barriers to applying big data into the smart factory	Big data analytics	Manufacturing	China	Review study	[75]
	Overview of Big data analytics in manufacturing IoT (MIoT)	Big data analytics	Manufacturing	China	Review study	[27]
	Use of Big data analytics for maintenance planning and fault pattern extraction	Big data analytics	Manufacturing and service	Australia	Special issue of journal	[123]
Integration, visualization, safety, and security	Implementation of fault detection and diagnosis in predictive maintenance	Big data analytics	Manufacturing	Australia	Experimental and descriptive	[163]

4.2.1.3 Digital twin The Digital Twin, resulting from advancements in the Industrial Internet of Things (IIoT), has evolved since its inception [44]. Digital Twin represents a unified virtual model of an actual work architecture that manages data related to physical and operational production firms [102]. Dubbed a "virtual factory," it synchronizes information and manages the units involved in the design, operation, and manufacturing of actual products. Its goal is to improve production and operational efficiency by fostering interconnection and assimilation through a real-world Industrial Internet of Things (IIoT) network [42, 114]. Table 9 outlines the investigations and emerging clusters utilizing the Digital Twin.

Employing ultra-realistic, dynamically adjusted simulation models, digital twin technology speeds up the product life cycle, addresses design and maintenance cost issues, and boosts manufacturing efficiency and quality [72, 156]. It overcomes cost and performance hurdles in personalized manufacturing and resolves integrated management issues in dispersed manufacturing. Features of the digital twin include hyperrealism, computability, controllability, and predictability [156]. The digital twin exhibits the following characteristics:

1. It mirrors the physical world in real-time, enabling digital twin applications to track and monitor numerous events in real-time.
2. It promotes growth through frequent interactions and high-level integration by communicating with diverse informational and functional components in both physical and digital domains. It offers integrations and value-adding opportunities, generating new analyses and accumulating information from both past and present data sources.
3. Even though similar to existing simulations, the digital twin performs additional functions through integrated analysis, such as facilitating rapid decision-making via the IIoT network and ensuring interoperability with other manufacturing applications [13].

4.2.1.4 Machine learning (ML) Machine learning (ML), a key aspect of artificial intelligence (AI), involves the automatic detection of significant patterns in large datasets. This technology enhances algorithm efficiency by enabling them to learn and adapt from big data analytics [122, 152]. It aids manufacturing organisations in uncovering new system patterns, correlating events (root cause analysis), predicting potential states, supporting data-driven decision-making, and optimising actions to improve manufacturing processes [160]. Table 10 summarises studies and emerging clusters in ML.

Machine learning technologies optimize, automate, and support complex problems that static, non-adaptive programs cannot solve [48]. For example, machine learning has been applied to predict bank credit ratings [74, 73]. A machine learning-based rescheduling framework addresses the increasing frequency and delays of rescheduling [74, 73]. A systematic literature review highlights the application of machine learning in intelligent logistics [152]. The findings identify seven clusters where machine learning is applicable: strategic and tactical process optimization,cyber-physical systems in logistics,predictive maintenance,hybrid

Table 9 Digital Twin based studies and emerging clusters

Industry 4.0 dimensions	Type of action taken	Digital technology	Industry	Country	Type of study	Author/year
Digital factory	Integrating CPS in digital factories	Digital twin	Manufacturing	China	Exploratory and descriptive	[132, 133]
Monitoring, tracking, and decision making	Connected micro smart factory	Digital twin, IIoT	Manufacturing	Korea	Case-based study	[102]
Interoperability and cost management	Implementing open-source technology in smart manufacturing	Digital twin	Manufacturing	Austria	Exploratory and descriptive	Damjanovic & Behrendt, [134]
	Rotating machinery fault diagnosis	Digital twin	Manufacturing	China	Experimental	[148]
Efficiency, flexibility, interconnection interoperability	Connecting physical shop floor and corresponding cyber-shop floor	Digital twin, CPS	Manufacturing	China	Exploratory and descriptive	[32]
Efficiency, flexibility	Job shop scheduling	Digital twin	Manufacturing	China	Experimental case study	[37]
Efficiency and flexibility	Robotics-based smart manufacturing systems	Digital twin,	Manufacturing	China	Exploratory and descriptive	[169]
Perception, interaction, automation, optimization	optimizing the physical and virtual layer	Digital twin	Manufacturing	China	Exploratory and descriptive	Vatankhah Barenji et al., [137]

Table 10 Digital Twin based studies and emerging clusters

Industry 4.0 dimensions	Type of action taken	Digital technology	Industry	Country	Type of study	Author/year
	Improving system performance	Deep learning algorithm	Manufacturing	China	Descriptive	[144]
	Job-shop Scheduling problem	AI Applications, deep Q network, edge computing	Semiconductor Manufacturing	Taiwan	Experimental	[78]
Productivity customization and efficiency	Production operating system	robotics and artificial intelligence	Small and medium-sized firms	Japan	Exploratory and descriptive	[151]
Reliability	Production System automated guided vehicles (AGVs	artificial intelligence, Mobile Intelligence	Manufacturing	Italy	Experimental and descriptive	[40]
	Cloud-to-edge based industrial cyber-physical systems	machine-learning techniques	Manufacturing	Spain	Exploratory and descriptive	[142]
Flexibility, reliability, waste management, reducing time	Job-shop scheduling	machine learning (ML) techniques	Plastic & rubber manufacturing	Italy	Experimental and descriptive	[74, 73]

decision support systems,production planning and control systems; and improving operational processes in logistics.

4.2.1.5 Cyber-physical system The cyber-physical system (CPS) is a prevalent digital technology that facilitates intelligent and autonomous interactions among production elements like assembly line machines, warehouses, supply chains, and customers, thereby transforming businesses into Smart Factories [93]. The term "cyber-physical system" was introduced by the United States National Science Foundation (NSF). CPS is a complex system comprising diverse components and structures tailored for various application scenarios [159].

CPS consists of four layers: physical, network, co-processing, and application layers [33]. It also includes a service-oriented architecture featuring sensor and actuator node modules, network modules, resource modules, and service modules. CPS addresses the limitations of computer-integrated manufacturing systems (CIMS), such as the lack of real-time data [168], insufficient information databases [11], and inadequate intelligence and proactivity [140].

CPS enables real-time data access, reconfiguration, interoperability, and decentralized decision-making, and is considered intelligent and proactive. A CPS-based smart manufacturing architecture comprises an input (customer requirements, materials, energy, capital, and labor), the Internet of Services (IoS) (connection, communication, interaction, and collaboration), the Internet of Things (IoT) (sensing and actuating), CPS (bridging the physical and cyber worlds), the Internet of Content Knowledge (IoCK), and a relationship (interactor) [159].

CPS bridges the physical and digital realms by integrating tools such as sensors, actuators, and collaborators into production devices. It provides manufacturing organizations with various services, including DaaS (Design as a Service), MFGaaS (Manufacturing as a Service), SaaS (Simulation as a Service), EaaS (Experimentation as a Service), MaaS (Management as a Service), MAaaS (Maintenance as a Service), and LaaS (Logistics as a Service). CPS also supports production modes like cloud manufacturing, wireless manufacturing, ubiquitous manufacturing, service-oriented manufacturing, and IPSS (Industrial Product Service System). Table 11 summarizes research on CPS and emerging clusters.

4.2.1.6 Cloud computing Cloud computing is an IT architectural model where hardware and software computations are abstracted and accessible online, on-demand, and independent of device or location [88]. It offers various services and deployment methods, distinguishing it from other technologies. Service models cater to customer expectations, including SaaS, PaaS, CaaS, IaaS, FaaS, and HaaS [105]. Table 12 summarizes studies and emerging clusters on cloud computing.

Cloud-based manufacturing is crucial in the service-oriented environment, simulation, and global services, gaining popularity among U.S. industrial enterprises [149]. Core features of cloud computing include resource abstraction, self-service, rapid elasticity, multi-tenancy, load balancing, and virtualization [155]. Business-relevant properties encompass quality of service, service level agreements, user

Table 11 Cyber-Physical System based studies and emerging clusters

Industry 4.0 dimensions	Type of action taken	Digital technology	Industry	Country	Type of study	Author/year
	Supply chain scheduling	CPS	Manufacturing	Germany	Empirical	[60]
	ICT into manufacturing	CPS	Manufacturing	USA	Empirical	[61]
Decision making	Solving types of scheduling problems	CPS	Manufacturing	Argentina	Exploratory & descriptive	[115]
Efficiency, cost-effectiveness, traceability, feasibility, capability	Returnable Transit Items, Logistics, and Distribution	CPS	Manufacturing	UK	Exploratory & descriptive	[98]
Interoperability, reliability, security, and privacy	Machine-to-machine (M2M) communication	Semantic-aware Cyber-Physical Systems (SCPSs)	Manufacturing	New Zealand	Exploratory and case-based study	[83]
Transparency, integration, and automation	Resource monitoring on the shop floor	CPS	Manufacturing	Australia	Exploratory & descriptive	[56]
Integration	Closed-loop adaptive scheduling solution	Cyber-Physical Production System (CPPS)	Manufacturing	China	Exploratory & descriptive	[111]

Table 12 Cyber-Physical System based studies and emerging clusters

Industry 4.0 dimensions	Type of action taken	Digital technology	Industry	Country	Type of study	Author/year
Integration, efficiency	Deployment of hierarchical fog server at the organizational network	Hierarchical fog computing	Bosch Electrical Manufacturing Company	France	Descriptive and Experimental	[18]
Connectivity mobility, energy management	Task offloading, path planning	Genetic algorithm, cloud robotics	Oil factory	Australia	Experimental	[113]
Time efficiency	Deploying IoT devices to ensure timely delivery of services	Fog computing	Computer Science	Australia	Descriptive and Experimental	[85]
Integration, cost reduction, improved efficiency	Enterprise resource planning (ERP) system	Cloud computing	Equipment and design manufacturers	China	Exploratory and Descriptive	[154]
Reduced wastage	Cloud-based Kanban decision support system	Cloud computing	Manufacturing	USA	Exploratory and review-based study	[121]

experience, fault tolerance, auditability, and certifiability [170]. Cloud technology enhances interoperability, deployment models, security, and business process management for enterprises and manufacturers [147]. Deployment strategies include public, private, community, and hybrid clouds.

Cloud manufacturing leverages cloud computing technology, integrating its architecture, tasks, and services into manufacturing companies [77, 153, 165]. Three manufacturing organizations utilize cloud computing for data gathering and analytics, inter-factory communication, and high-performance computing and simulation. SaaS and FaaS in cloud computing streamline manufacturing processes. The primary barriers to cloud adoption are interoperability and data portability, not security [150], due to the rapid increase of Internet-connected devices in new business models [36].

4.2.1.7 Blockchain technology "Blockchain technology" denotes an extensive database supported by advanced cryptography and a consensus structure [57, 58]. It consists of transactions combined into a single block. Blockchain technologies help maintain consensus on the current state and progression of shared facts [129]. There are two types: chain-structured and DAG-structured blockchains [65]. Most blockchain applications, such as Bitcoin, Ethereum, and Hyperledger, use chain-structured blockchains. However, its complex cryptographic security makes it energy-intensive, unsuitable for power-constrained IoT technology. An alternative, known as tangle, is based on a directed acyclic graph design [109].

Blockchain technology has only recently been applied to IoT due to the latter's architectural limitations in handling an increasing number of nodes and expanding network sizes [143]. As more IoT applications are deployed, the number of connected devices rises, generating significant data that complicates meeting quality-of-service requirements (QoS). Blockchain technology has thus emerged as a solution to IoT's shortcomings.

Blockchain allows for the decentralized storage and management of data, ensuring its trustworthiness and confidentiality. This decentralization eliminates failures, facilitates the integration and tracking of devices, and creates a resilient ecosystem, leading to cost savings for IoT manufacturers. Cryptographic algorithms within blockchain technology guarantee customer data privacy and confidentiality. Blockchain enables decentralized data storage, transparency, integrity, and resilient networking. It has been widely used in financial markets to analyze the relationship between Bitcoin (BP) and oil (OP) prices, and manufacturing groups have adopted a blockchain reference architecture [171]. Table 13 summarizes studies and emerging clusters involving Blockchain Technology.

5 I4.0 and DI new trends in addition to themes

The section examines emerging trends and themes in I4.0 and DI literature through six sub-sections.

Table 13 Blockchain Technology based studies and emerging clusters

Industry 4.0 dimensions	Type of action taken	Digital technology	Industry	Country	Type of study	Author/year
Energy management	Blockchain architecture for IoT devices	Blockchain	Computer Science	China	Case study	[58, 57]
Security and privacy	Blockchain architecture to reshape traditional IIoT,	Blockchain	Computer Science	China	Case study	[143]
	To evaluate the impact of Bitcoin price (BP) and oil price (OP) upon each other	Blockchain	Financial Markets	China	Exploratory and descriptive	[128]
Reliability, efficiency, automation	Blockchain reference architecture	Blockchain	Manufacturing	USA	Exploratory and descriptive	[171]

5.1 Emerging research trends in the fields of industry 4.0 and digital innovation

Here, we conclude our investigation by addressing our third research question: What are the research gaps, constraints, and recommendations for future scholars and practitioners in "Industry 4.0" and "Digital Innovation"? While extensively studied, Industry 4.0 remains nascent. Originating in Germany, it spread to the U.S. and other economies as part of national strategies for the fourth industrial revolution, achieved through digital innovations like Big Data, Cloud computing, CPS, Machine Learning, and IoT. Post-COVID-19, various sectors adopted digital technologies to maintain operations, but Industry 4.0 is mostly limited to manufacturing. Within manufacturing, MNEs predominantly implement it, whereas SMEs are still exploring its features.

Challenges to adopting Industry 4.0 include insufficient human resources and suboptimal work conditions, financial constraints, standardisation issues, cybersecurity and data ownership concerns, risk of fragility, technological integration, coordination challenges across organisational units, and plausibility issues [55]. These barriers affect both MNEs and SMEs.

Our investigation revealed that Industry 4.0 is primarily restricted to manufacturing. However, research needs to extend to other sectors like hospitality and tourism, healthcare, banking and finance, education, and other service industries. Each industry has unique characteristics and organisational systems. Without adequate guidance or readiness assessments, adopting digital innovation for Industry 4.0 is risky. Thus, companies must learn to manage change before embarking on digital innovation.

Businesses can align their organizational strategy with Industry 4.0's characteristics. For instance, China declared the fourth industrial revolution through the "Made in China 2025" strategy, providing technical and financial support to research institutes and manufacturing enterprises. Similarly, the European Commission's Digital Innovation Hub (DIH) programs assist companies in digital innovation by enabling digital technology testing and pilot project experimentation for universities, research centers, private companies, incubators, and start-up accelerators [52]. The Indian government is also launching schemes and policies to promote digital innovation across various sectors, including agriculture, retail, renewable energy, health products, electric vehicles, and technology centers. By facilitating technological, financial, and behavioral support, the Indian government aims to transform SMEs and MSEs in rural and urban areas for enhanced innovation and sustainable growth (Kadaba, Aithal, & KRS, [124]).

5.2 Future research on industry 4.0 and digital innovation

Recent attention from scholars, academicians, and practitioners has been directed towards Industry 4.0 and digital innovation [15, 47, 124]. The growing volume of research articles has led to more case-based, conceptual, and systematic reviews. Additionally, this study outlines future research opportunities in Industry 4.0 and digital innovation, detailed in Table 14.

Table 14 Summary of research questions and future research opportunities

No	Themes of the studies	Research questions and future research work	Future research suggestions
1	Industry 4.0	RQ1: What is the current status of Industry 4.0 in both developing and developed economies? RQ2: What is the short, medium, and long term impact of Industry 4.0 on organizational sustainability? RQ3: How Industry 4.0 dimensions can be successfully implemented in different sectors of the economy? RQ4: What are the inhibitors or exhibitors of Industry 4.0? RQ5: How organizations can transform their business models for implementing Industry 4.0? RQ6: What are the challenges faced by managers while implementing Industry 4.0 in large, medium, or small-sized organizations? RQ7: What is the impact of Industry 4.0 on the dynamically changing business requirements?	*Methodology:* Future researchers can work on empirical and case-based studies *Context:* (a) Comparative studies among developing and developed economies (b) Impact of using digital technologies in different sectors other than manufacturing (c) Using diversified contexts in the area of Industry 4.0 (d) Evaluating the impact of Industry 4.0 while considering social-economic factors
2	Smart manufacturing	RQ1: What is the impact of Industry 4.0 in transforming traditional manufacturing into smart manufacturing? RQ2: How Industry 4.0 can be functional in manufacturing organizations? RQ3: How can organizations modify their structures to implement smart manufacturing? RQ4: How to implement smart manufacturing models into large, medium, and small enterprises? RQ5: What are the inhibitors and exhibitors of implementing smart manufacturing in both developing and developed economies? RQ6: How does smart manufacturing helps in attaining organizational sustainability?	

Table 14 (continued)

No	Themes of the studies	Research questions and future research work	Future research suggestions
3	Digital transformation	RQ1: What is the role of Industry 4.0 in the digital transformation of an organization? RQ2: How to handle digital transformation in large, medium, and small enterprises? RQ3: What challenges are organizations facing while taking the route to digital transformation? RQ4: What are the key dimensions of digital transformation? RQ5: How organizations can be transformed digitally? RQ6: How digital transformation can help in attaining organizational sustainability?	
4	ABCD technologies (artificial intelligence, block chain, cloud computing and data Analytics)	RQ1: Which of the recent ABCD technologies are adopted by organizations in both developing and developed economies? RQ2: Which of the factors affect the adoption and implementation of ABCD technologies in organizations? RQ3: Which of the enablers facilitate the implementation of ABCD technologies in small, medium, and large organizations? RQ4: How do ABCD technologies operate to transform organizations into smart factories? RQ5: What are the opportunities and challenges of ABCD technologies? RQ6: How do ABCD technologies help in attaining long-term sustainability?	

This review rigorously examines literature to identify existing themes in the relationship between I4.0 and DI. It defines I4.0, its dimensions, along with its significance for organizational sustainability, and explores the digital technologies organizations employ to evolve into smart manufacturing entities.

5.3 Industry 4.0

Industry 4.0 or I4.0 addresses evolving consumers' anticipations by using digitally innovative tools and techniques such as Big Data, Blockchain, Cloud computing, Digital Twin, IoT, and ML (AI) to guarantee on-time and in-budget facilitation and delivery of goods and services. Extensive research exists on Industry 4.0, but future case-based or empirical studies could further explore its impact in both developing and developed economies. Comparative studies on organizational culture, structure, and geographical, social, and economic factors can expand research in this area.

5.4 Smart manufacturing

Smart manufacturing provides several advantages to customers, the environment, and the economy by digitising its processes. These are as follows:

With digitization companies provide personalized approaches to their customers.
Companies can maintain the demand and supply chain which reduces their operational costs.
Digitization has transformed the processes into automation which further reduced production time.
Digitization and I4.0 improves organizational competence by monitoring and controlling the processes while saving resources and operational costs.
Digitization reduces manual labour, personnel errors, and attains efficiency in services.
Digitization also enables easy sharing of data and information among employees.
More personalization helps manufacturers to personally and efficiently address customers' demands.
DI and Industry 4.0 focusses on attaining efficiency in using energy. It helps in producing goods and services which use less energy for a sustainable environment.
Production of energy efficient products will capture market share as they are energy efficient and satisfy environmental conscious customers.

Previous studies have addressed supply chain management [146], job-shop scheduling [56, 158], enterprise resource planning, virtual manufacturing [23], agile manufacturing [45], and lean manufacturing [107]. Research has particularly focused on Italy, Germany, France, the United Kingdom, and other European countries that have recently invested in smart manufacturing. Future research should include a broader sample of companies from various countries and sectors to confirm earlier findings, using both qualitative (e.g., case studies) and quantitative (e.g., survey) methods. Additionally, future studies could examine variables such as country, industry, and dimensions (or strategies and techniques) of smart manufacturing.

5.5 Digital transformation

Previous research explores principles, patterns, contexts, and convergence requirements for transforming manufacturing units into smart manufacturing ones, focusing on resource virtualization and cloud computing. It also addresses the complexities of big data analytics and machine learning. Future research should investigate the challenges of implementing digital technologies like big data analytics, machine learning, and blockchain, considering technological, societal, environmental, and technical factors. Comparative studies among small, medium, and large organizations undergoing digital transformation are needed. Researchers should also conduct qualitative and quantitative studies to examine the pre-and post-impact of digital transformation on organizations.

5.6 ABCD technologies

This study identifies blockchain technology as an emerging field attracting researcher attention. It also examines ABCD technologies—artificial intelligence, blockchain, cloud computing, and data analytics—in manufacturing organizations. These technologies' applications in other economic sectors can be analyzed through empirical, conceptual, or case-based studies. This research contributes to Industry 4.0 and digital innovation knowledge, aiding future scholars and practitioners. Initially aimed at creating smart factories, Industry 4.0 has been mainly studied in manufacturing.

The authors plan to explore other economic sectors related to Industry 4.0 and assess its impact on job growth in emerging economies like India, China, Malaysia, Bangladesh, Thailand, and the Philippines. Future research should investigate Industry 4.0's role in sustainable development by reducing emissions and waste and promoting green policies. Additionally, studies should focus on digital innovation-based startups and government-supported ventures to achieve Industry 4.0 goals, including "Digital India," "Self-Reliant India" (Atma Nirbhar Bharat), and "Make in India."

6 Managerial implications of the study

Infusing digital innovation to attain Industry 4.0 dimensions can significantly impact upon organizational operations, strategy, and competitive positioning. Organizations' investments in new technologies such as IoT, AI, big data, and robotics helps managers to develop a clear vision and plan long-term strategies to attain organizational objectives. These technologies also ensure efficient allocation of resources and offer considerable return on their implementation. These digital innovations radically transforms organizations' rigid hierarchical structures into flexible ones. They also cultivate a culture that embraces change, encourages continuous learning, and supports innovation which is also crucial to attain organizational sustainability. Moreover, implementing digital technologies helps managers in identifying the actual areas of training and development to upskill or reskill their workforce. It helps in talent retention, skill development of workforce, and their performance measurement while ensuring full transparency to the employees with these advanced technologies.

Managers must identify digital technologies like automation and integration which streamline operations, reduce costs, and improve operational efficiency of organizational processes. Similarly, leveraging big data and analytics develop managerial capabilities to collect, analyse, and real-time decision-making for organizational performance. By impanelling continuous digital innovation, managers can create such environments that support experimentation, intrapreneurship, engagement with external partners, and a collaborative ecosystem. Adopting new technologies introduces new risks, including cybersecurity threats for which managers must implement robust risk management frameworks and ensure compliance with regulations. To achieve Industry 4.0, managers must implement comprehensive changes in management practices, adopting a proactive, strategic, and adaptive approach to fully leverage digital technologies for sustained growth and competitiveness.

7 Conclusions and limitations of the study

The current study was initiated with the objective to examine the themes depicting the association between DI and I4.0, which envisions a smarter, more automated, and efficient future for products, services, and industries. Industry 4.0 characteristics, such as automation, integration, flexibility, collaboration, virtualization, safety, and security, can be achieved through the implementation of smart technologies like Big Data Analytics, Cloud Computing, Cyber-Physical Systems, Digital Twins, IoT, Machine Learning (AI), and Blockchain technology, which drive changes across various economic sectors (agriculture, education, medicine, pharmaceutical, automobile, real estate, etc.).

The research, based on a literature survey of articles from the Web of Science database (2011–2021), classified digital technologies according to key themes, characteristics, implementation patterns, advantages, and downsides, and summarized their roles in achieving Industry 4.0's dimensions. The study expands the understanding of digital tools and technologies (artificial intelligence, IoT, and big data) used extensively in economies for attaining sustainable competitive advantage, although limited by the chosen database and time period.

Another perspective of DI and I4.0 is that they require only professional, skilled, and trained employees. It reduces the labour force and increases unemployment. A minor error or technical issue can affect the whole production process. Manufacturers depend upon technologies and machines to complete their orders while limiting their options and variables. Companies will be unable to face uncertainties or unseen issues that can occur due to changing political and global scenarios.

I4.0, or Industry 4.0, has been recently introduced but cannot be adopted fully by both developed and developing economies. Developing economies have several economic and structural loopholes or shortcomings while adopting Industry 4.0 dimensions. Industrial 4.0 and Digital Innovation are a threat to environmental sustainability. Implementation of digital technologies requires skilled and professional employees, which are in short supply compared to demand.

Studies have suggested that DI technologies are important to transform traditional companies into smart factories [87]. However, it is equally important to have a strong research and development center, an incubation and innovation centre, and government support to infuse these digital technologies into different sectors of the economy. Thus, there is a need for more experimental and empirical research from scientists, academicians, and researchers to support this change. The transformation of manufacturing companies into smart factories can bring more business opportunities and the digital world can grow 10 times more than in the current scenario. However, DI is based on data security, user privacy, and cyber security, which can be attained only by consistency and scalability.

Small, medium, and large companies are focusing on attaining sustainability and progress by implementing DI technologies, but it is not easy due to internal and external factors such as political, geographical, economic, infrastructural, legal, environmental, and social. Also, it is difficult for companies to adopt and implement DI technologies because it increases their dependence on technologies, and these technologies do not have an emotional or psychological understanding of human emotions [138]. DI has been implemented in sectors like manufacturing, retail, supply chain and logistics, pharmaceuticals, agriculture, automobiles, etc., to attain Industry 4.0, but it has a negative impact on the health and development of human beings as they become less creative and innovative. For instance, AI-based ChatGPT is used for finding answers to most of the questions, which limits the critical thinking ability of individuals as they believe and limit themselves to the information provided by the tool. Thus, DI helps in attaining the dimensions of industry 4.0 but it also has its own limitations and loopholes which need to be addressed in the futuristic research studies.

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